

By Proposition 3, we have

$$\begin{aligned}
 \text{var}_{p \times q} \left(\widehat{\text{MMD}}_{\lambda}^2(\mathbb{X}_m, \mathbb{Y}_n) \right) &\leq C_1(M, d) \left(\frac{\|\psi * \varphi_{\lambda}\|_2^2}{m+n} + \frac{1}{(m+n)^2 \lambda_1 \cdots \lambda_d} \right) \\
 2\sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_{\lambda}^2(\mathbb{X}_m, \mathbb{Y}_n) \right)} &\leq 2\sqrt{\frac{2C_1}{\beta} \frac{\|\psi * \varphi_{\lambda}\|_2^2}{m+n} + \frac{2C_1}{\beta(m+n)^2 \lambda_1 \cdots \lambda_d}} \\
 &\leq 2\sqrt{\|\psi * \varphi_{\lambda}\|_2^2 \frac{2C_1}{\beta(m+n)} + \frac{2\sqrt{2C_1}}{\sqrt{\beta(m+n)} \sqrt{\lambda_1 \cdots \lambda_d}}} \\
 &\leq \|\psi * \varphi_{\lambda}\|_2^2 + \frac{2C_1}{\beta(m+n)} + \frac{2\sqrt{2C_1}}{\sqrt{\beta(m+n)} \sqrt{\lambda_1 \cdots \lambda_d}} \\
 &\leq \|\psi * \varphi_{\lambda}\|_2^2 + \frac{2C_1 + 2\sqrt{2C_1}}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \ln\left(\frac{1}{\alpha}\right) \\
 &\leq \|\psi * \varphi_{\lambda}\|_2^2 + \frac{6C_1}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \ln\left(\frac{1}{\alpha}\right) \\
 \|\psi * \varphi_{\lambda}\|_2^2 - 2\sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_{\lambda}^2 \right)} &\geq -6C_1 \frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}}
 \end{aligned}$$

where for the third inequality we used the fact that $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ for all $x, y > 0$, for the fourth inequality we used the fact that $2\sqrt{xy} \leq x+y$ for all $x, y > 0$, and for the fifth inequality we use the fact that $\lambda_1 \cdots \lambda_d \leq 1$, $\beta \in (0, 1)$ and $\ln\left(\frac{1}{\alpha}\right) > 1$. A similar reasoning has been used by Fromont et al. (2013, Theorem 1) and Albert et al. (2022, Theorem 1).

Let $C_3(M, d) := 6C_1(M, d) + 2\sqrt{2}C_2(M, d)$ where C_1 and C_2 are the constants from Propositions 3 and 4, respectively. Assume that our condition holds, that is

$$\|\psi\|_2^2 - \|\psi - \psi * \varphi_{\lambda}\|_2^2 \geq (2\sqrt{2}C_2 + 6C_1) \frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}}.$$

Omitting the variables for $\hat{q}_{1-\alpha}^{\lambda, B}(\mathbb{Z}_B | \mathbb{X}_m, \mathbb{Y}_n)$ and for $\widehat{\text{MMD}}_{\lambda}^2(\mathbb{X}_m, \mathbb{Y}_n)$, we then get

$$\begin{aligned}
 &\mathbb{P}_{p \times q \times r} \left(2\hat{q}_{1-\alpha}^{\lambda, B} \leq \|\psi\|_2^2 - \|\psi - \psi * \varphi_{\lambda}\|_2^2 + \|\psi * \varphi_{\lambda}\|_2^2 - 2\sqrt{\frac{2}{\beta} \text{var}_{p \times q} \left(\widehat{\text{MMD}}_{\lambda}^2 \right)} \right) \\
 &\geq \mathbb{P}_{p \times q \times r} \left(2\hat{q}_{1-\alpha}^{\lambda, B} \leq (6C_1 + 2\sqrt{2}C_2) \frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} - 6C_1 \frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \\
 &= \mathbb{P}_{p \times q \times r} \left(\hat{q}_{1-\alpha}^{\lambda, B} \leq \sqrt{2}C_2 \frac{\ln\left(\frac{1}{\alpha}\right)}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \\
 &\geq \mathbb{P}_{p \times q \times r} \left(\hat{q}_{1-\alpha}^{\lambda, B} \leq C_2 \sqrt{\frac{2}{\beta(m+n) \sqrt{\lambda_1 \cdots \lambda_d}}} \ln\left(\frac{1}{\alpha}\right) \right) \\
 &\geq 1 - \frac{\beta}{2}
 \end{aligned}$$

where the third inequality holds because $\beta \in (0, 1)$ and the last one holds by Proposition 4 since $B \geq \frac{3}{\alpha^2} \left(\ln \left(\frac{8}{\beta} \right) + \alpha(1 - \alpha) \right)$. Lemma 2 then implies that

$$\mathbb{P}_{p \times q \times r} \left(\Delta_{\alpha}^{\lambda, B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 0 \right) \leq \beta.$$

E.6 Proof of Theorem 6

Theorem 5 gives us a condition on $\|\psi\|_2^2 - \|\psi - \psi * \varphi_{\lambda}\|_2^2$ to control the power of the test $\Delta_{\alpha}^{\lambda, B}$. We now want to upper bound $\|\psi - \psi * \varphi_{\lambda}\|_2^2$ in terms of the bandwidths when assuming that the difference of the densities lie in a Sobolev ball. We first prove that if $\psi := p - q \in \mathcal{S}_d^s(R)$ for some $s > 0$ and $R > 0$, then there exists some $S \in (0, 1)$ such that

$$\|\psi - \psi * \varphi_{\lambda}\|_2^2 - S^2 \|\psi\|_2^2 \leq C'_4(d, s, R) \sum_{i=1}^d \lambda_i^{2s} \quad (24)$$

for some positive constant $C'_4(d, s, R)$.

For $j = 1, \dots, d$, since $K_j \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, it follows by the Riemann-Lebesgue Lemma that its Fourier transform $\widehat{K}_j$ is continuous. For $j = 1, \dots, d$, note that

$$\widehat{K}_j(0) = \int_{\mathbb{R}} K_j(x) e^{-ix \cdot 0} dx = \int_{\mathbb{R}} K_j(x) dx = 1$$

and, since $K_j \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, also that

$$\prod_{j=1}^d |\widehat{K}_j(\xi_j)| \leq \prod_{j=1}^d \int_{\mathbb{R}} |K_j(x) e^{-ix \cdot \xi_j}| dx = \prod_{j=1}^d \int_{\mathbb{R}} |K_j(x)| dx =: \kappa_1 < \infty$$

as defined in Equation (21). We deduce that $\left| 1 - \prod_{i=1}^d \widehat{K}_i(\xi_i) \right| \leq 1 + \kappa_1$ for all $\xi \in \mathbb{R}^d$. Let us define $g: \mathbb{R}^d \rightarrow \mathbb{R}$ by $g(\xi) = 1 - \prod_{i=1}^d \widehat{K}_i(\xi_i)$ for $\xi \in \mathbb{R}^d$. We have $g(0, \dots, 0) = 0$, so by continuity of g , there exists some $t > 0$ such that

$$S := \sup_{\|\xi\|_2 \leq t} |g(\xi)| < 1.$$

For any $s > 0$, we also define

$$T_s := \sup_{\|\xi\|_2 > t} \frac{\left| 1 - \prod_{i=1}^d \widehat{K}_i(\xi_i) \right|}{\|\xi\|_2^s} \leq \frac{1 + \kappa_1}{t^s} < \infty.$$

Let $\Psi := \psi - \psi * \varphi_{\lambda}$. As it is a scaled product of $K_1, \dots, K_d \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, we have $\varphi_{\lambda} \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. Since we assume that $\psi \in \mathcal{S}_d^s(R)$, we have $\psi \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. For $p \in \{1, 2\}$, since $\psi \in L^1(\mathbb{R}^d)$, we have $\|\psi * \varphi_{\lambda}\|_p \leq \|\psi\|_1 \|\varphi_{\lambda}\|_p < \infty$. Hence, we deduce that $\Psi \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. By Plancherel's Theorem, we then have

$$\begin{aligned} (2\pi)^d \|\Psi\|_2^2 &= \|\widehat{\Psi}\|_2^2 \\ (2\pi)^d \|\psi - \psi * \varphi_{\lambda}\|_2^2 &= \left\| (1 - \widehat{\varphi_{\lambda}}) \widehat{\psi} \right\|_2^2. \end{aligned}$$

In general, for $a > 0$ the Fourier transform of a function $x \mapsto \frac{1}{a}f(\frac{x}{a})$ is $\xi \mapsto \widehat{f}(a\xi)$. Since $\varphi_\lambda(u) := \prod_{i=1}^d \frac{1}{\lambda_i} K_i\left(\frac{u_i}{\lambda_i}\right)$ for $u \in \mathbb{R}^d$, we deduce that $\widehat{\varphi_\lambda}(\xi) = \prod_{i=1}^d \widehat{K}_i(\lambda_i \xi_i)$ for $\xi \in \mathbb{R}^d$. Therefore, we have

$$\begin{aligned}
 & (2\pi)^d \|\psi - \psi * \varphi_\lambda\|_2^2 \\
 &= \left\| (1 - \widehat{\varphi_\lambda}) \widehat{\psi} \right\|_2^2 \\
 &= \int_{\mathbb{R}^d} |1 - \widehat{\varphi_\lambda}(\xi)|^2 |\widehat{\psi}(\xi)|^2 d\xi \\
 &= \int_{\mathbb{R}^d} \left| 1 - \prod_{i=1}^d \widehat{K}_i(\lambda_i \xi_i) \right|^2 |\widehat{\psi}(\xi)|^2 d\xi \\
 &= \int_{\|\xi\|_2 \leq t} \left| 1 - \prod_{i=1}^d \widehat{K}_i(\lambda_i \xi_i) \right|^2 |\widehat{\psi}(\xi)|^2 d\xi + \int_{\|\xi\|_2 > t} \left| 1 - \prod_{i=1}^d \widehat{K}_i(\lambda_i \xi_i) \right|^2 |\widehat{\psi}(\xi)|^2 d\xi \\
 &\leq S^2 \int_{\|\xi\|_2 \leq t} |\widehat{\psi}(\xi)|^2 d\xi + T_s^2 \int_{\|\xi\|_2 > t} \|(\lambda_1 \xi_1, \dots, \lambda_d \xi_d)\|_2^{2s} |\widehat{\psi}(\xi)|^2 d\xi \\
 &\leq S^2 \|\widehat{\psi}\|_2^2 + T_s^2 \int_{\mathbb{R}^d} \left(\sum_{i=1}^d \lambda_i^2 \xi_i^2 \right)^s |\widehat{\psi}(\xi)|^2 d\xi \\
 &\leq S^2 (2\pi)^d \|\psi\|_2^2 + T_s^2 \int_{\mathbb{R}^d} \left(\sum_{i=1}^d \lambda_i^2 \right)^s \left(\sum_{i=1}^d \xi_i^2 \right)^s |\widehat{\psi}(\xi)|^2 d\xi \\
 &= S^2 (2\pi)^d \|\psi\|_2^2 + T_s^2 \|\lambda\|_2^{2s} \int_{\mathbb{R}^d} \|\xi\|_2^{2s} |\widehat{\psi}(\xi)|^2 d\xi \\
 &\leq S^2 (2\pi)^d \|\psi\|_2^2 + T_s^2 \|\lambda\|_2^{2s} (2\pi)^d R^2
 \end{aligned}$$

since $\psi \in \mathcal{S}_d^s(R)$, and where we have used Plancherel's Theorem for $\psi \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. We have proved that there exists some $S \in (0, 1)$ such that

$$\|\psi - \psi * \varphi_\lambda\|_2^2 \leq S^2 \|\psi\|_2^2 + T_s^2 R^2 \|\lambda\|_2^{2s}.$$

If $s \geq 1$, then $x \mapsto x^s$ is convex and, by Jensen's inequality (finite form), we have

$$\|\lambda\|_2^{2s} = \left(\sum_{i=1}^d \lambda_i^2 \right)^s = d^s \left(\sum_{i=1}^d \frac{1}{d} \lambda_i^2 \right)^s \leq d^s \sum_{i=1}^d \frac{1}{d} (\lambda_i^2)^s = d^{s-1} \sum_{i=1}^d \lambda_i^{2s} \leq d^{1+s} \sum_{i=1}^d \lambda_i^{2s}.$$

If $s < 1$, then $\gamma := \frac{1}{s} > 1$ and so, it is a standard result that $\|\cdot\|_\gamma \leq \|\cdot\|_1$. We then have

$$\|\lambda\|_2^{2s} = \left(\sum_{i=1}^d \lambda_i^2 \right)^s = \left(\sum_{i=1}^d (\lambda_i^{2s})^\gamma \right)^{1/\gamma} = \|\lambda^{2s}\|_\gamma \leq \|\lambda^{2s}\|_1 = \sum_{i=1}^d \lambda_i^{2s} \leq d^{1+s} \sum_{i=1}^d \lambda_i^{2s}.$$

Hence, for all $s > 0$, we have $\|\lambda\|_2^{2s} \leq d^{1+s} \sum_{i=1}^d \lambda_i^{2s}$. We conclude that

$$\|\psi - \psi * \varphi_\lambda\|_2^2 \leq S^2 \|\psi\|_2^2 + T_s^2 R^2 d^{1+s} \sum_{i=1}^d \lambda_i^{2s}$$